

STATEMENT OF RESEARCH INTERESTS

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The Problem

AI safety research remains disproportionately centered on English. Many widely used safety benchmarks, red-teaming protocols, and alignment methods were developed primarily using English-language data and evaluated within English-speaking contexts. This creates an important gap for languages whose linguistic variation and cultural context are not adequately represented in existing evaluation frameworks.

Arabic illustrates this challenge particularly well. Spoken across more than 20 countries, Arabic encompasses Modern Standard Arabic (MSA) alongside diverse regional varieties, including Gulf, Levantine, Egyptian, and Maghrebi Arabic. These varieties differ substantially in vocabulary, syntax, pragmatics, and culturally specific expressions. Harmful intent can also be conveyed through dialectal expressions, euphemisms, coded language, sarcasm, and contextual references that models trained predominantly on MSA or translated English data may fail to recognize.

My research program addresses this gap by developing **Arabic-first, dialect-aware, and multimodal approaches to trustworthy AI and online safety**. Rather than treating Arabic as a translated extension of English NLP, my work builds datasets, evaluation frameworks, models, and research infrastructure grounded in the linguistic and socio-cultural characteristics of Arabic digital discourse.

What I Have Built

Over the past several years, I have developed an integrated body of work spanning **dataset construction, benchmark design, multimodal modeling, shared-task organization, and deployed NLP infrastructure**.

Population-Scale Social Media Analysis

The **MARSAD Social Media Observatory** is a large-scale platform I helped develop at HBKU for collecting and analyzing social media discourse. It integrates data from platforms including X/Twitter, Telegram, Reddit, YouTube, and Bluesky and supports Arabic NLP capabilities such as sentiment analysis, propaganda detection, fact-checking signals, claim analysis, hateful-content detection, topic analysis, and network-level exploration.

Developing MARSAD has connected my methodological research with practical system-building. It has required addressing challenges including scalable transformer inference, multilingual and dialectal processing, multimodal analysis of image-text content, and the translation of model outputs into interpretable analytical interfaces. This experience has shaped an important principle underlying my research: NLP systems for online safety should be evaluated not only as classifiers, but also as components of larger human-centered analytical workflows.

Datasets and Benchmarks

Reliable Arabic safety research requires evaluation resources that reflect the linguistic diversity of Arabic. I have therefore invested substantially in constructing datasets and benchmarks rather than relying primarily on translated English resources.

The **So Hateful!** dataset (LREC-COLING 2024) provides fine-grained annotations of Arabic hate speech and has received **40 Google Scholar citations since its 2024 publication**, indicating growing reuse as a

community resource. My more recent work, **ArabicDialectSafety**, extends safety evaluation across six Arabic varieties and examines both binary harmful-content detection and fine-grained categories of harm. This work demonstrates the importance of evaluating safety across dialects rather than treating Arabic as a homogeneous language.

Together, these projects support a broader objective: developing evaluation resources that reveal **where Arabic language models fail, which linguistic varieties are affected, and what kinds of harmful content remain difficult to detect**.

Multimodal Arabic NLP

Online harm is not exclusively textual. Memes and other image-text combinations can communicate hate, propaganda, political framing, and implicit meaning through interactions between visual and linguistic cues.

My work on **Arabic multimodal propaganda and harmful-content detection**, including ArAIEval 2024, MAHED 2025, ImageEval 2025, and AHA-Memes 2026, investigates vision-language approaches for these settings. I have worked with multimodal architectures including CLIP-based systems and recent vision-language models such as LLaMA-Vision and Qwen-VL, incorporating OCR, textual context, and visual representations.

A central challenge is that multimodal meaning is often culturally situated. Political symbols, regional references, visual stereotypes, dialectal expressions, and historical context may determine whether an image-text combination is harmful. Consequently, effective Arabic multimodal safety systems require culturally grounded datasets and evaluation protocols rather than straightforward transfer from English-language meme benchmarks.

Community Infrastructure Through Shared Tasks

A complementary part of my research has been building community evaluation infrastructure. I have organized or co-organized international shared tasks including **ArGuard 2026, MAHED 2025, StanceNakba 2026, ArAIEval, CheckThat!, and ImageEval**.

Shared tasks provide something individual benchmark papers cannot easily achieve: a common dataset, standardized evaluation protocol, competitive baselines, and a community examining the same research problem from different methodological perspectives. Through these initiatives, I aim to make Arabic NLP resources reusable beyond an individual publication and to encourage reproducible comparison across systems.

This combination of datasets, benchmarks, systems, and community resources is reflected in my broader publication record. My Google Scholar profile currently reports **862 citations, an h-index of 14, and an i10-index of 19**. Importantly, **834 citations (96.8%) have accumulated since 2021**. My citation record has also begun shifting from earlier work in health informatics and medical AI toward Arabic NLP: for example, *So Hateful!* (2024) has received 40 citations, while several of my recent Arabic NLP and multimodal publications have already reached double-digit citations.

The Research Agenda: Toward Interpretable, Context-Aware Arabic AI Safety

The datasets and systems I have developed expose an important limitation of current content-safety models. A classifier may correctly identify harmful content while providing little insight into **why** the content is harmful, which contextual signals produced the decision, or whether that reasoning remains valid across dialects and communities.

This becomes particularly problematic when harmful intent is expressed indirectly-through sarcasm, coded expressions, euphemisms, visual references, conversational context, or coordinated behavior across multiple messages.

The next phase of my research therefore moves from **detecting harmful content toward understanding and explaining harm**. I plan to pursue this agenda through three interconnected directions.

1. Interpretable Content-Safety Models

Most existing Arabic hate-speech, propaganda, and safety classifiers provide predictions without explanations that correspond to concepts meaningful to human analysts. I aim to develop models that jointly predict harmful-content categories and produce structured evidence explaining those decisions.

Rather than relying solely on post-hoc token attribution methods, I am interested in **concept-level interpretability**: identifying the dialectal expression, rhetorical strategy, visual symbol, target group, or contextual relationship that supports a prediction and connecting that evidence to an explicit annotation taxonomy.

Such systems would allow researchers and domain experts to inspect disagreements between model predictions and human interpretations. More importantly, explanation quality itself can become an evaluation target rather than an optional visualization layered on top of classification accuracy.

2. Dialogue- and Context-Aware Harm Detection

Most content-moderation benchmarks treat messages independently. Real harmful behavior often does not occur this way.

Harassment can emerge through repeated interactions; coordinated campaigns distribute narratives across accounts; implicit hate may become identifiable only from earlier conversational context; and multi-turn jailbreaks can construct harmful requests incrementally even when individual turns appear benign.

I plan to develop **conversation- and sequence-level models** that represent these interactions explicitly. The objective is to model how stance, sentiment, targets, and intent evolve across messages and determine when individually ambiguous content forms a harmful discourse pattern collectively.

My recent work on stance detection and multi-task modeling of Arabic discourse provides an initial foundation for this direction. Longer term, I am interested in extending the unit of safety analysis from the **utterance to the conversation, and eventually from the conversation to interaction networks**.

3. Cross-Dialectal Robustness and Low-Resource Transfer

Arabic safety models must also operate under substantial linguistic distribution shift. A model performing well on MSA or Egyptian Arabic cannot automatically be assumed to behave similarly on Moroccan, Algerian, Palestinian, Syrian, or other Arabic varieties.

Collecting equally large annotated datasets for every variety is unlikely to be feasible. I therefore plan to investigate **cross-dialect transfer, controlled dialectal augmentation, representation alignment, and calibration under distribution shift**. A particular question is whether safety-relevant concepts can be represented independently from dialect-specific surface forms while preserving culturally specific distinctions that should not be normalized away.

This direction would extend my existing dialectal safety work from documenting performance disparities toward developing methods that systematically reduce them.

These three directions share a common objective: **Arabic AI safety systems should be accurate, robust, and legible**. Their decisions should be understandable to researchers evaluating them, practitioners deploying them, and communities affected by them.

Teaching and Mentorship

My teaching and mentoring interests closely complement this research agenda. I have taught **Principles of Computing and Information Management Systems** and have mentored students in research methodology, annotation design, machine-learning experimentation, and academic writing.

I view research mentorship as training students to conduct reproducible research independently: formulating testable questions, constructing reliable datasets, designing appropriate baselines and ablations, analyzing failure cases, and communicating results clearly.

I am particularly interested in mentoring students working on Arabic NLP, multilingual AI, multimodal learning, and responsible AI. Expanding the community of researchers who understand both modern NLP methodology and the linguistic realities of Arabic is essential for developing technologies that adequately serve Arabic-speaking communities.

Fit and Collaboration

My research lies at the intersection of **natural language processing, multimodal AI, trustworthy AI, and computational social science**. This creates natural opportunities for collaboration with researchers studying multilingual and low-resource NLP, AI safety, vision-language models, misinformation and online harm, computational social science, and human-centered AI.

I bring experience across the full research pipeline—from collecting and annotating data to developing models, organizing community evaluations, and deploying analytical systems. My future goal is to build on this foundation by developing **interpretable, context-aware, and dialectally robust AI systems for Arabic**, while using Arabic as a challenging testbed for broader questions about multilingual AI safety.

Ultimately, I want my research to contribute to a shift from asking whether a model is *safe in Arabic* to asking a more demanding question: **Does the model understand enough about the language, context, and community to explain when and why something is harmful?**